

Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

AIM: To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Objective:

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

2.

To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

3.

To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

Objective 1 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Throughput (Mbps)
```

```
NB_IoT = 0.05*(1-exp(-SNR/8));
```

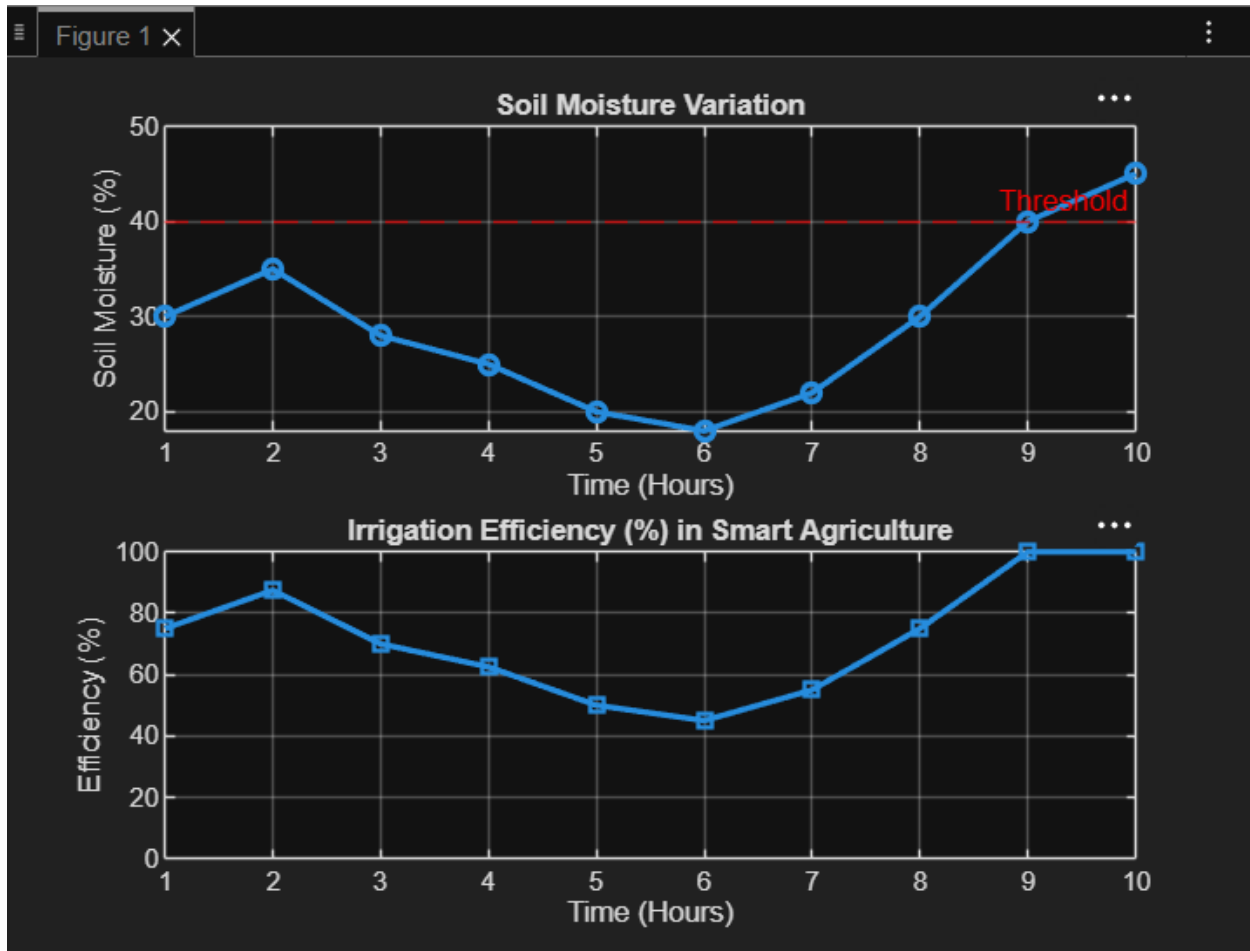
```
LTE_M = 1.2*(1-exp(-SNR/8));
```

```
disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
```

```
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----  
subplot(2,1,1)  
plot(SNR,NB_IoT,'-o','LineWidth',2)  
title('NB-IoT Throughput vs SNR')  
xlabel('Signal-to-Noise Ratio (dB)')  
ylabel('Throughput (Mbps)')  
grid on  
  
% ----- Graph 2 -----  
subplot(2,1,2)  
plot(SNR,LTE_M,'-s','LineWidth',2)  
  
title('LTE-M Throughput vs SNR')  
xlabel('Signal-to-Noise Ratio (dB)')  
ylabel('Throughput (Mbps)')  
grid on
```



The throughput of both NB-IoT and LTE-M increases as the Signal-to-Noise Ratio (SNR) improves. LTE-M achieves significantly higher throughput than NB-IoT, making it more suitable for applications requiring higher data rates.

Objective 2 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Latency (ms)
```

```
NB_IoT = [180 160 145 130 120 110 100];
```

```
LTE_M = [90 80 70 60 50 45 40];
```

```
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
```

```
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR,NB_IoT,'-o','LineWidth',2)
```

```
title('NB-IoT Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

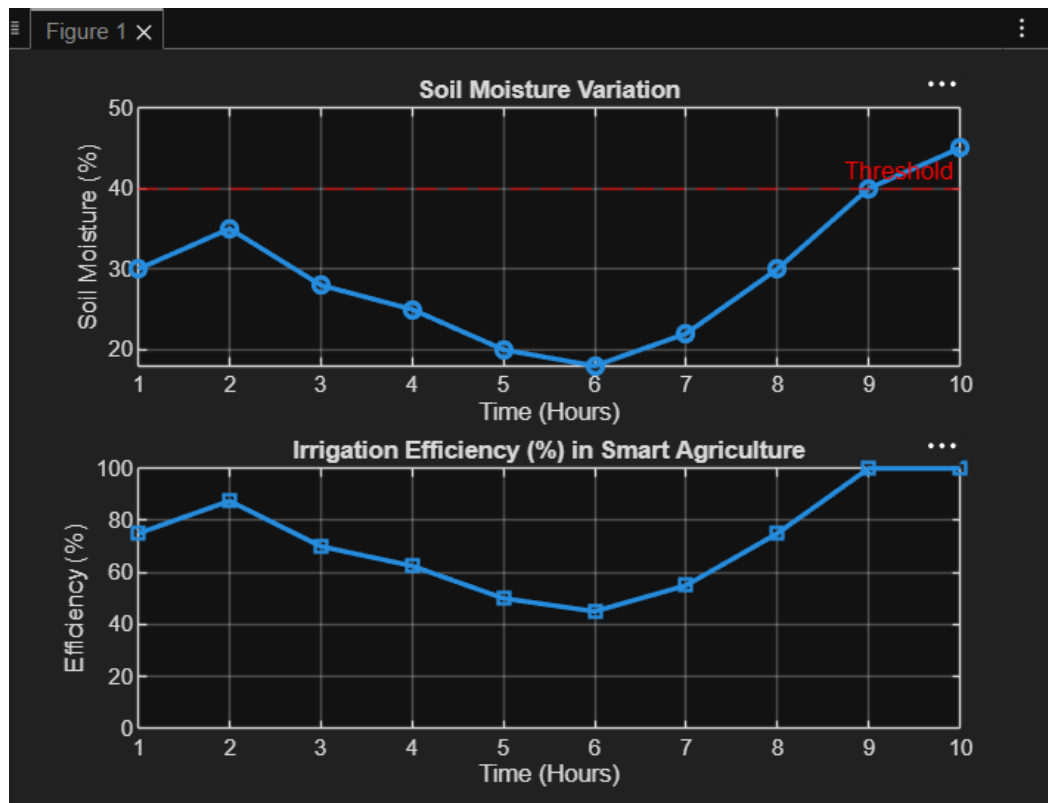
```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

```
title('LTE-M Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```



The latency of both NB-IoT and LTE-M decreases as the SNR increases due to improved channel quality. LTE-M exhibits lower latency than NB-IoT, making it more suitable for delay sensitive IoT applications such as real-time monitoring and asset tracking. Objective 3 Matlab

Code and Output:

```
clc;
clear;
close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];

disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%))')
disp([SNR' NB_IoT' LTE_M'])
```

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

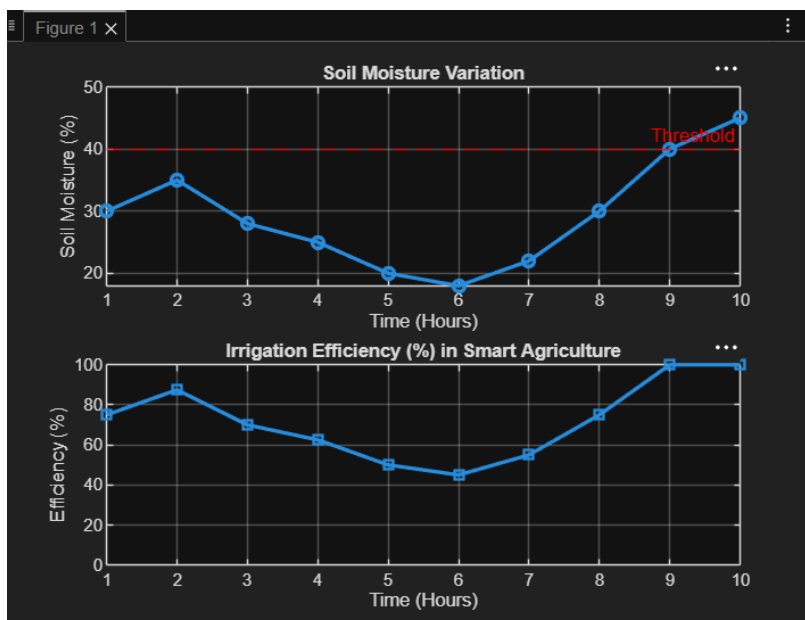
plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on



The Packet Delivery Ratio (PDR) of both NB-IoT and LTE-M increases as the Signal-to-Noise

Ratio (SNR) improves. LTE-M achieves a slightly higher PDR than NB-IoT, indicating more reliable packet transmission under the same channel conditions.

Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions. 2. The latency analysis showed that LTE-M provides lower communication delay compared to

NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating